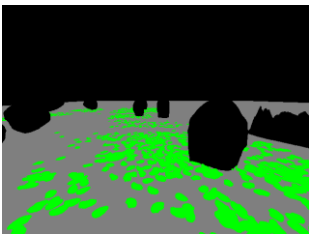


Collect Images

Segmentations(Seg)



Depth Image(Dt)



Perception

Occupancy Map Update

3D Point Cloud
Generation

Point Cloud
Segmentation

2D Map
Projection

Map Update

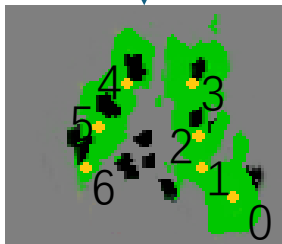
Occupancy Map

Centroid Analysis

Centroid
Extract

Centroid cross
frame track

Centroid Chain



Hierarchical Planner

High Level Planner

VLM

Task Related
Information

Chain of
thought

Enable

Replan

Result

Smart Trigger

Arrive?

Stuck?

Low Level Planner
Fail?

Geometric Verifier

Deviate
?

Turn
Back?

Low Level Planner

MDDP

Generate Speed
Pair

Sample
Trajectory

Cost Calculation

Final Trajectory

— Constant Information Flow

..... Triggered information flow

O_c : Occupancy map with centroid

G: Global Waypoint